M2 Internship Report

Change-point detection algorithms for marked Poisson processes and application for automated heat detection in small ruminants

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Context

This internship was carried out within the framework of the SynchroBIO project (2025–2026), an exploratory research project supported by INRAE's METABIO metaprogramme, which focuses on the scaling-up of organic agriculture. SynchroBIO is dedicated to the automated detection of oestrus and to the organization of artificial insemination after hormone-free synchronization by the male effect in small ruminants.

The internship took place from March 2026 to July 2026 at the UMR MISTEA (Mathematics, Computer Science and Statistics for the Environment and Agronomy), on the La Gaillarde campus of the Institut Agro Montpellier. It was supervised by Bertrand Cloez, Nicolas Verzelen and Isabelle Sanchez from UMR MISTEA.

The work focused on the development of statistical methods for change-point detection in marked Poisson processes, with an application to the automatic detection of oestrus in small ruminants. More precisely, we assume that the observed behaviours can be represented by a marked Poisson process. The aim is then to detect a point at which the intensity or the parameters of this process change. This change-point is interpreted as the time at which the animal changes its behaviour. It can therefore be used to predict the beginning of the period in which the animal is in heat.

This report presents the statistical framework studied during the internship, the proposed change-point detection procedures, and their application to this biological and agricultural problem.

Chapter 1

Introduction

1.1 Background and Model

Change-point detection is a classical problem in statistics, with applications in many fields such as climate change detection [16], medical condition monitoring [21], speech recognition, image analysis, and human activity analysis [3]. Its aim is to identify time instants at which the probabilistic behaviour of an observed process changes. Depending on the application, the change may affect the mean of a time series, the intensity of a counting process, the distribution of additional marks, or several of these quantities simultaneously. The literature usually distinguishes between single and multiple change-point detection, between offline and online settings, and between discrete-time and continuous-time observations [8, 13, 20].

In the offline setting, the whole observation interval is available before the statistical decision is made. The objective is then to decide whether one or several changes occurred during the experiment, and to estimate their locations. In contrast, in the online setting, observations are collected sequentially and the procedure must be updated as time goes on. The aim is to raise an alarm as soon as possible after a change, while controlling the risk of false alarm. This sequential point of view goes back to the cumulative sum procedure introduced in [4], and remains a central framework for modern online change-point detection [4, 17].

In this internship, we develop statistical algorithms to address the problem raised in the SynchroBIO project, whose objective is to design digital tools for the automated detection of oestrus in small ruminants. The data considered in this work consist of interaction events between males and females. For each female, we observe a sequence of event times together with additional information attached to each event, such as the duration of the

interaction. Such observations are naturally represented as marked point processes. More precisely, for an individual i, the data observed on a time interval $[0, T]$ are written as $\{(T_{ij}, U_{ij}) : j = 1, \ldots, N_i(T)\}$, where T_{ij} is the time of the j-th observed interaction, U_{ij} is the associated mark, and $N_i(T)$ is the total number of observed events before time T.

A simple statistical model assumes that each individual can be in one of two states: a pre-oestrus state (pre-change state) and an oestrus state (post-change state). The transition time between these two states is unknown and is denoted by S_i. Before S_i, the events are modelled by a marked Poisson process with parameters (λ_0, μ_0), whereas after S_i they are modelled by another marked Poisson process with parameters (λ_1, μ_1). Here, λ represents the intensity of the point process, while μ is related to the distribution of the marks. Thus, the onset of oestrus may be associated with a change in the rate at which events occur, and possibly with a change in the distribution of the marks. This leads to a change-point detection problem for marked Poisson processes, a framework closely related to recent work on offline segmentation of inhomogeneous and marked Poisson processes [8].

In this report, we consider both offline and online versions of the problem. In the offline setting, an existing likelihood-based algorithm had already been developed by Bertrand Cloez, Isabelle Sanchez and Marie-Pierre Etienne (ENSAI Rennes). During this internship, this existing offline procedure was studied and improved by incorporating information related to the different males involved in the interactions. In addition, a new offline algorithm was explored and compared with the existing one, in order to assess whether it could improve the detection results or provide a more suitable approach for the available data. We also constructed new online approaches, which are closer to the final objective of an alert system: at each time, only past observations can be used, and an alarm must be raised when there is enough evidence that the behaviour of an individual has changed.

In this internship, several individuals are observed simultaneously. For each individual, we want to test whether a change in behaviour has occurred during the observation period. Therefore, the same statistical test has to be applied several times, once for each individual. This leads to a multiple testing difficulty: controlling the error of each test separately is not sufficient to control the error of the whole procedure [11]. Indeed, even if each individual test has a small probability of producing a false detection, performing many tests at the same time increases the probability of obtaining at least one false detection somewhere in the herd. Thus, the relevant error is no longer only the error of one test, but the global error induced by the collection of tests.

For this reason, the detections cannot be validated by applying a classical single-test rule independently to each female. The problem must be formu-

lated as a multiple testing problem. In this internship, the error control criterion is chosen according to the detection algorithm and the statistical objective. In particular, in the offline setting, the false discovery rate (FDR) is used when the aim is to control the expected proportion of false detections among all detected individuals [5]. By contrast, in the online setting, family-wise error rate (FWER) control [11] is used to control the probability of raising at least one false alarm over all monitored individuals and all monitoring times.

1.2 Outline

The remainder of this report is organized as follows. Chapter 2 presents the agronomic context of the study and describes the available datasets. Chapter 3 introduces the offline change-point detection procedures, where the whole observation period is available before making a decision. Chapter 4 is devoted to the online setting, where observations are processed sequentially and alarms must be raised in real time. Chapter 5 applies the proposed methods to simulated and experimental data, and compares the offline and online results. Finally, Chapter 6 summarizes the main conclusions of the internship and discusses possible perspectives for future work.

Chapter 2

Agronomic Context and Data Description

This chapter presents the agronomic and biological context and describe the available datasets during the internship.

2.1 Agronomic and biological context

In sheep and goat farming systems, the control of reproduction is an important issue for organizing mating and artificial insemination. In conventional breeding systems, hormonal treatments are commonly used to induce or synchronize oestrus, because they make it easier to group reproductive events and to plan insemination. However, in organic farming, reproductive management is expected to rely mainly on natural methods. Although artificial insemination remains allowed, hormonal treatments cannot be used as a routine tool for controlling reproduction [9]. This restriction is consistent with the general objectives of organic farming, which include reducing external inputs, respecting animal welfare, and maintaining consumer trust in organic products. Therefore, alternative hormone-free strategies are needed for reproductive management in organic sheep and goat farming systems.

One of these alternatives is synchronization by the male effect, a natural technique that can induce off-season ovulation and support mating or artificial insemination in small ruminants [6]. The principle is to use the presence of males as a biological stimulus for the females. After a period with little or no contact with males, their reintroduction can stimulate reproductive activity in the females and trigger oestrus and ovulation without using hormones. The male effect is therefore useful because it can increase the number of females entering oestrus within a relatively short period. Once females are

in oestrus, artificial insemination can be organized. However, insemination must be performed at the right period with respect to the onset of oestrus. Detecting this period at the right time is essential in order to inseminate females under appropriate conditions. Oestrus is associated with changes in female behaviour, especially in their interactions with males. Around the onset of oestrus, females may become more receptive to males, and the frequency of mounting events may increase. These behavioural changes provide useful information for identifying the reproductive state of each female.

In this study, the available data come from the Alpha-Detector electronic oestrus detection system [1,2], which records mounting events between males and females. This system relies on RFID technology: the male is equipped with a harness carrying an RFID reader, while the female carries an RFID transponder. When the male mounts a female, the reader is triggered and identifies the female through her transponder. The system then records the identity of the ewe, the date and time of the mount, and additional information related to the interaction, such as its duration. These recorded events provide indirect information on sexual behaviour and can therefore be used to study and detect oestrus .

From an agronomic point of view, the objective is to transform these behavioural records into reliable decision-support information for farmers and experimental teams. More precisely, the aim is to detect the onset of oestrus automatically after hormone-free synchronization by the male effect, in order to help organize mating or artificial insemination in small ruminants [6]. This biological and agronomic motivation leads naturally to a statistical change-point detection problem: for each female, we observe a sequence of interaction events over time, and we want to identify whether and when a change in behaviour occurs.

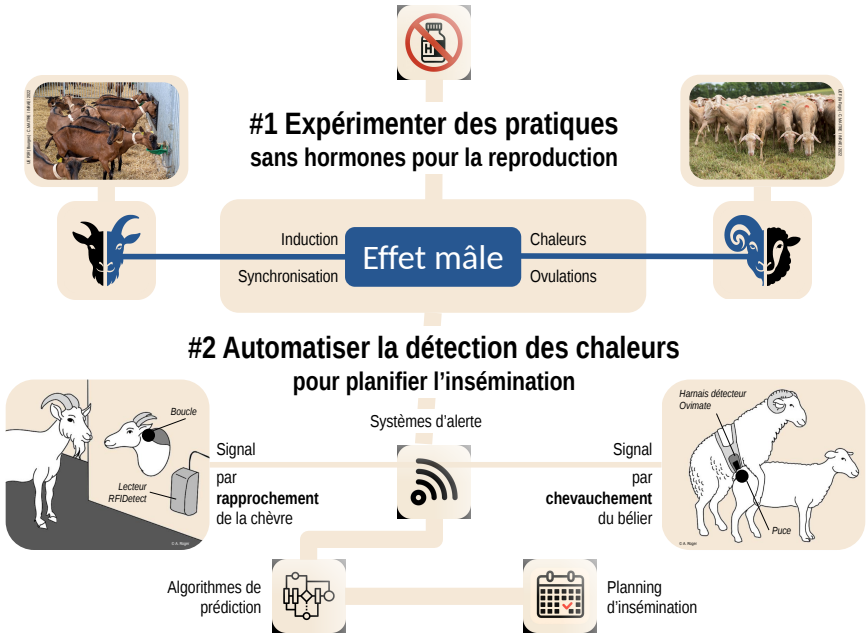


Figure 2.1: General procedure of the SynchroBIO project.

2.2 Data Description

In this study, the available data contain records of interactions between males and females, together with complementary information about the experimental protocols and artificial insemination. A dataset from 2014 (epic 2014) had already been used to test the existing offline detection algorithm. During this internship, additional datasets were made available from several years and from three different geographical and experimental contexts: La Fage, La Cazotte, and Corse. These sites correspond to different farming conditions and animal populations, including different sheep breeds.

The new datasets include protocol files, artificial insemination data, male-effect data, and Alpha-Detector records, with the date and time of the detected mounting events, the duration of the interactions, the identifiers of the animals involved, and other complementary information. They cover different sites and years: La Fage 2023 and 2025, La Cazotte 2023 and 2024, and Corse 2024 and 2025. This diversity makes it possible to test the detection procedures in several real situations and animal populations.

However, discussions with Sandrine Fréret and Nathalie Debus, who provided the data and with whom we worked directly during the internship,

indicated that the recorded interaction durations were not fully reliable. For this reason, during the data preparation and analysis, we avoided using the duration variable as a main source of information.

In addition, we also used an empirical manual classification rule associated with the Alpha-Detector protocol. This rule is based on the cumulative number of mounting events recorded for each ewe within a day. If this number is smaller than 3, the ewe is considered as showing no clear behavioural sign of heat. If it lies between 3 and 5, the situation is considered uncertain. If it is at least 6, the ewe is considered as showing a strong behavioural signal of heat, that is, a strong signal suggesting that the ewe is in heat. This rule does not estimate a change-point time; it only provides a practical manual criterion against which the rupture decisions of the detection algorithms can be compared.

Overview of the datasets

Before preprocessing, the data were provided as raw Excel files. These files contained one row for each recorded event, with information such as the animal identifiers, the male identifier, the date and time of the event, cumulative counts, and other variables produced during the experiment. Figure 2.2 shows an example of the raw structure of the data before cleaning.

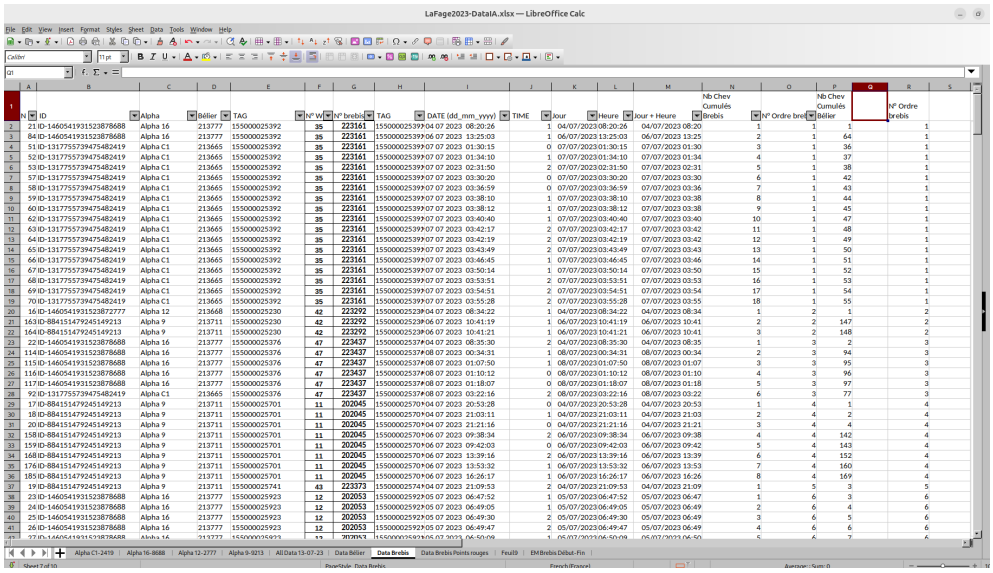


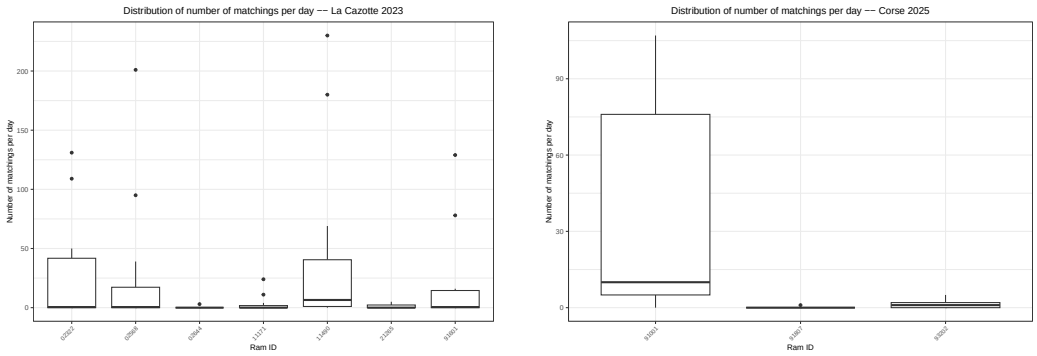
Figure 2.2: Example of raw data structure in an Excel file before preprocessing.

To make the data easier to use in R, we kept only the variables needed for the analysis were selected from the raw Excel files. The cleaned datasets

were then converted into `.rda` files and work with this in R.

Dataset	Ewes	Rams	Period	Days
Epic 2014	153	7	17/04/2014–15/05/2014	28
Corse 2024	31	1	21/05/2024–24/05/2024	4
Corse 2025	26	3	20/05/2025–24/05/2025	5
La Fage 2023	32	4	04/07/2023–08/07/2023	5
La Fage 2025	44	4	13/06/2025–27/06/2025	15
La Cazotte 2023	78	7	09/05/2023–20/05/2023	12
La Cazotte 2024	83	5	21/05/2024–02/06/2024	13

Table 2.1: Overview of the datasets



(a) La Cazotte 2023 (b) Corse 2025

Figure 2.3: Distribution of the number of matchings per day for each ram.

Figure 2.3 shows that the number of matchings per day is not homogeneous across rams. Some rams have higher activity levels than others, suggesting that the male effect may vary from one ram to another.

Most rams were different across sites and years. Only two rams were observed in two different years within the same site. Ram ID 11171 was recorded in La Cazotte 2023 and La Cazotte 2024, while ram ID 93202 was recorded in Corse 2024 and Corse 2025. These rams can be used to check whether the Poisson intensity associated with the male effect remains stable across years within the same site.

The observation duration also differs across datasets. Some datasets cover only a few days, whereas others contain a longer monitoring period. This should be considered when applying the online detection algorithms discussed in Chapter 4. In particular, the alarm rule may be adapted to the length of

the experiment by using a finer or coarser monitoring grid, for example by checking the alarm condition every few hours or once per day.

Chapter 3

Offline Change-Point Detection Algorithm

This chapter introduces the offline change-point detection models. We first present the existing offline algorithm and its adaptation, in particular the addition of a ram-effect parameter to account for differences between males. We then describe the new offline algorithm developed in this work and compare it with the existing approach on the available data.

As discussed in Chapter 1, several ewes are tested simultaneously. Therefore, the error of the whole multiple-testing procedure has to be controlled. In the offline framework, we control the false discovery rate (FDR) using the Benjamini–Hochberg procedure [5].

The algorithms considered in this report are designed to detect at most one change-point for each ewe. During this internship, I also explored an extension to the case of two change-points, in order to detect both the beginning and the end of heat. However, this approach was not retained, because I did not find a satisfactory testing procedure to distinguish between zero, one, and two change-points.

3.1 Existing Offline Likelihood-Based Algorithm with Ram Effect

The offline algorithm considered in this section is based on a likelihood approach previously developed by Bertrand Cloez, Isabelle Sanchez and Marie-Pierre Etienne (ENSAI Rennes). In its initial form, the same pre-change and post-change parameters were used for the whole dataset.

During this internship, the algorithm was adapted in order to include the effect of the ram involved in each mounting event. The objective is to allow

each ram to have its own parameters before and after the change-point. This extension accounts for the fact that different males may have different levels of activity. The principle of the algorithm is to combine a bi-optimization procedure with a multiple-testing step.

3.1.1 Model with ram-specific parameters

We consider n independent marked Poisson processes observed on the time interval $[0, T]$. For process i, the observations are denoted by

$$X_i = \{(T_{ij}, U_{ij}, R_{ij}) : j = 1, \dots, N_i\},$$

where T_{ij} is the time of the j-th event, U_{ij} is the associated mark, $R_{ij} \in \mathcal{R}$ is the ram involved in the event and N_i is the total number of observed events for process i on $[0, T]$.

Each process i may have an unknown change-point S_i. For a given ram r, the intensity is assumed to be piecewise constant and the marks are assumed to follow exponential distributions

$$\lambda_r(t) = \begin{cases} \lambda_r^{(0)}, & t \le S_i, \\ \lambda_r^{(1)}, & t > S_i. \end{cases} \qquad U_{ij} \sim \begin{cases} \text{Exp}(\mu_r^{(0)}), & T_{ij} \le S_i, \\ \text{Exp}(\mu_r^{(1)}), & T_{ij} > S_i, \end{cases} \qquad r = R_{ij}.$$

For a given change-point S_i, under this model, the log-likelihood of the whole dataset is

$$\begin{aligned} \ell(\mathbf{S}, \theta) = \sum_{i=1}^{n} \ell_i(S_i, \theta) = \sum_{i=1}^{n} \sum_{r=1}^{R} \Big[& (\lambda_r^{(1)} - \lambda_r^{(0)}) S_i - \lambda_r^{(1)} T + N_{ir}^{(0)} \log(\lambda_r^{(0)} \mu_r^{(0)}) \\ & + N_{ir}^{(1)} \log(\lambda_r^{(1)} \mu_r^{(1)}) - \mu_r^{(0)} V_{ir}^{(0)} - \mu_r^{(1)} V_{ir}^{(1)} \Big]. \end{aligned} \tag{3.1}$$

where $\mathbf{S} = (S_1, \dots, S_n)$ and $\theta = (\lambda^{(0)}, \lambda^{(1)}, \mu^{(0)}, \mu^{(1)}) = \{\lambda_r^{(0)}, \lambda_r^{(1)}, \mu_r^{(0)}, \mu_r^{(1)} : r = 1, \dots, R\}$. The objective is to estimate both the change-points $\mathbf{S}$ and the ram-specific parameters θ.

For the remainder of this report, all mathematical calculations and algorithms adhere to the specific notations and conventions detailed in Appendix C.

3.1.2 Bi-optimization algorithm

The parameters and the change-points have to be estimated by maximizing the log-likelihood. For fixed parameters $\theta = (\lambda^{(0)}, \lambda^{(1)}, \mu^{(0)}, \mu^{(1)})$, the derivative of the log-likelihood with respect to S is

$$\frac{\partial}{\partial S} \ell(\mathbf{S}, \theta) = -\lambda^{(0)} + \lambda^{(1)}$$

Between two consecutive observed event times, the quantities $N_i^{(0)}$, $N_i^{(1)}$, $V_i^{(0)}$ and $V_i^{(1)}$ remain constant. Therefore, as a function of the candidate change-point S, the log-likelihood is affine. Consequently, its maximum on each interval is attained at one of the endpoints. Since the only points where the counts and the sums of marks can change are the observed event times, it is sufficient to search over the finite candidate set $\mathcal{G}_i$.

Thus, for fixed θ, the estimated change-point is defined by

$$\widehat{S}_i(\theta) \in \arg\max_{S \in \mathcal{G}_i} \ell_i(S; \theta).$$

Figure 3.1 illustrates this idea. The log-likelihood is affine on each open interval between two observed event times and may be discontinuous at the observed event times. Its maximum is therefore reached at one of the observed event times.

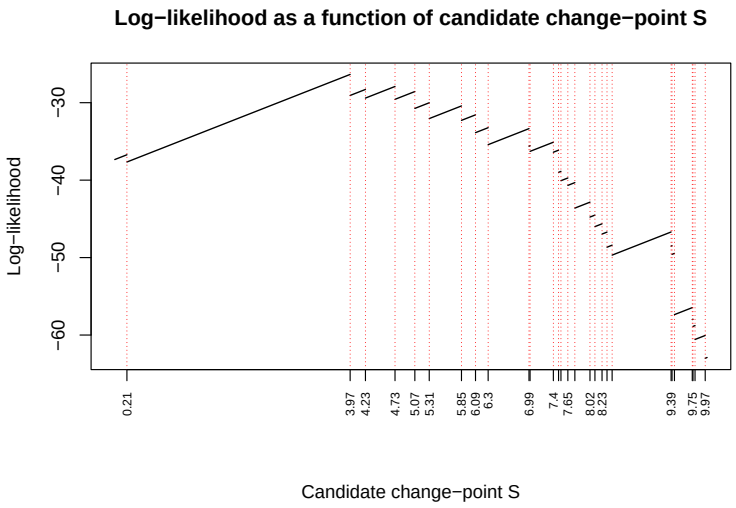


Figure 3.1: Illustration of the log-likelihood in Equation (3.1) as a function of the candidate change-point S, for fixed parameters $\lambda = (1, 4)$ and $\mu = (2, 1)$.

Conversely, for fixed change-points $S_1, \ldots, S_n$, the maximum likelihood estimators of the ram-specific parameters are

$$\widehat{\lambda}_r^{(0)} = \frac{\sum_{i=1}^n N_{ir}^{(0)}}{\sum_{i=1}^n S_i}, \quad \widehat{\lambda}_r^{(1)} = \frac{\sum_{i=1}^n N_{ir}^{(1)}}{\sum_{i=1}^n (T - S_i)}, \quad \widehat{\mu}_r^{(0)} = \frac{\sum_{i=1}^n N_{ir}^{(0)}}{\sum_{i=1}^n V_{ir}^{(0)}}, \quad \widehat{\mu}_r^{(1)} = \frac{\sum_{i=1}^n N_{ir}^{(1)}}{\sum_{i=1}^n V_{ir}^{(1)}}.$$

These formulas show that the estimate $\hat{\theta}$ depends on the current values of the change-points. Conversely, the estimated change-points depend

on the current parameter values. For this reason, we use the bi-optimization algorithm, where the estimation procedure alternates between two steps: updating the change-points for fixed parameters and updating the parameters for fixed change-points.

3.1.3 Multiple testing step

Moreover, not every process necessarily undergoes a change-point, so an estimated candidate change-point does not always correspond to a true change. Therefore, a testing step is added to decide which processs have a statistically significant change-point. For each process i, we test

$$\mathcal{H}_0^{(i)} : \text{no change-point occurs on } [0, T], \text{ or } S_i = T$$

against

$$\mathcal{H}_1^{(i)} : \text{one change-point occurs on } [0, T].$$

Likelihood-ratio statistic.

Let $\ell_i^{(0)} = \ell_i(T; \widehat{\theta})$ denote the log-likelihood contribution of process i under the null hypothesis, and let $\ell_i^{(1)} = \max_{S \in \mathcal{G}_i\}} \ell_i(S; \widehat{\theta})$ denote the maximum log-likelihood contribution under the alternative hypothesis. The likelihood-ratio-type statistic is then defined by

$$D_i = \ell_i^{(1)} - \ell_i^{(0)}.$$

Large values of D_i indicate evidence against the null hypothesis and in favor of the presence of a change-point.

Parametric bootstrap.

The null distribution of D_i is not available in closed form. We therefore approximate it using a parametric bootstrap procedure [10].

For this purpose, we generate M_{boot} trajectories under the null model, that is, under a marked Poisson process with parameters estimated from the data and the change-point is fixed at T.

For each simulated trajectory, we compute the corresponding statistic $D_i^{(1)}, \ldots, D_i^{(M_{\text{boot}})}$. The bootstrap p-value associated with process i is then estimated by

$$p_i = \frac{1}{M_{\text{boot}}} \sum_{b=1}^{M_{\text{boot}}} \mathbf{1}\{D_i^{(b)} > D_i\}.$$

Benjamini–Hochberg procedure.

Since the test is performed simultaneously on n process, we correct for multiple comparisons using the Benjamini–Hochberg procedure [5] in order to control the false discovery rate.

Let $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(n)}$ denote the ordered p-values. For the level α, we define

$$k = \max \left\{ i \in \{1, \ldots, n\} : p_{(i)} \leq \frac{i}{n} \alpha \right\}.$$

The null hypotheses corresponding to $p_{(1)}, \ldots, p_{(k)}$ are rejected, and the associated individuals are declared to have a change-point. For the remaining individuals, the null hypothesis is not rejected and their change-point is set to $S_i = T$.

The complete procedure then

Algorithm 1 Offline likelihood-based detection with ram effect

1: **Input:** Event data $X_1, \ldots, X_n$, T, M_{boot}, α, and maximum number of iterations $M_{\max}$.
2: **Output:** Final change-points $S_1^{(M_{\max})}, \ldots, S_n^{(M_{\max})}$ and estimated parameters.
3: Initialize $\lambda_r^{(0)}$, $\lambda_r^{(1)}$, $\mu_r^{(0)}$ and $\mu_r^{(1)}$ for all rams $r = 1, \ldots, R$.
4: **for** $m = 1, \ldots, M_{\max}$ **do**
5: **Change-point update:** for each process i, compute

$$\widehat{S}_i^{(m)} = \in \arg\max_{S \in \mathcal{G}_i} \ell_i(S; \theta^{(m-1)}).$$

6: **Testing step:** use the likelihood-ratio statistic, the parametric bootstrap and the Benjamini–Hochberg procedure to decide whether $S_i^{(m)} = \widehat{S}_i^{(m)}$ or $S_i^{(m)} = T$.
7: **Parameter update:** using $S_1^{(m)}, \ldots, S_n^{(m)}$, update $\lambda_r^{(0,m)}$, $\lambda_r^{(1,m)}$, $\mu_r^{(0,m)}$ and $\mu_r^{(1,m)}$ for each ram r.
8: **end for**
9: Return $S_1^{(M_{\max})}, \ldots, S_n^{(M_{\max})}$ and the final parameter estimates.

3.2 New Offline Algorithm: Selection–Testing Procedure

In Algorithm 1, the null distribution of the test statistic is approximated by Monte Carlo simulations. Although this approach is flexible, it can be computationally expensive and depends on the number of simulated samples. In this section, we present a new procedure in which the distribution of the test statistic can be approximated theoretically under suitable asymptotic assumptions. This procedure is inspired by the idea of selection and testing for change-point detection in point processes [8].

The main idea is to split the observed marked Poisson process into two independent parts. The first part is used to select candidate change-points, whereas the second part is used to test whether these candidates correspond to significant changes.

Remark. The selection–testing procedure presented in this section is written without a ram effect. A ram-effect version could be developed in the same spirit by estimating separate parameters for each ram. In the present work, we keep the procedure in its simpler form without ram effect in order to validate the principle of the method.

The procedure contains three main steps: data splitting by thinning, selection of candidate change-points on the selection sample, and multiple testing on the test sample.

Data splitting by thinning.

For each event of each process, we generate an independent Bernoulli random variable

$$B_{ij} \sim \text{Bernoulli}(q), \qquad q \in (0,1).$$

We define

$$B_{ij} = 1 \quad \Longrightarrow \quad (T_{ij}, U_{ij}) \in \mathcal{D}^{\text{sel}},$$

and

$$B_{ij} = 0 \quad \Longrightarrow \quad (T_{ij}, U_{ij}) \in \mathcal{D}^{\text{test}}.$$

Thus,

$$\mathcal{D}^{\text{sel}} = \{(T_{ij}, U_{ij}) : B_{ij} = 1\}, \qquad \mathcal{D}^{\text{test}} = \{(T_{ij}, U_{ij}) : B_{ij} = 0\}. \tag{3.2}$$

By the thinning property of Poisson processes, the two samples $\mathcal{D}^{\text{sel}}$ and $\mathcal{D}^{\text{test}}$ are independent. If the original intensity in state C is $\lambda^{(C)}$, then the selection sample has intensity $a^{(C)} = q\lambda^{(C)}$, whereas the test sample has intensity $b^{(C)} = (1-q)\lambda^{(C)}$. Only the event intensity is split between the selection and test samples; the distribution of the marks remains the same.

Selection step.

The selection sample $\mathcal{D}^{\text{sel}}$ is used to estimate candidate change-points. On this sample, we run the bi-optimization procedure of Algorithm 1, without the testing step and without ram-specific parameters. After M_{sel} iterations, we obtain the candidate change-points $S_i^{\text{cand}} = S_i^{(M_{\text{sel}})}$, and the estimated parameters on the selection sample, $a = (a^{(0)}, a^{(1)}), \quad \mu = (\mu^{(0)}, \mu^{(1)})$.

Testing step.

The testing step is performed on $\mathcal{D}^{\text{test}}$, independently of the selection step. For each process i, the candidate change-point $S_i = S_i^{\text{cand}}$ is kept fixed. We test

$$\mathcal{H}_0^{(i)} : S_i = T \qquad \text{against} \qquad \mathcal{H}_1^{(i)} : S_i = S_i^{\text{cand}}.$$

Under $\mathcal{H}_1^{(i)}$, the maximum likelihood estimators on the test sample are

$$\widehat{b}_i^{(0)} = \frac{N_{i,\text{test}}^{(0)}}{S_i}, \qquad \widehat{b}_i^{(1)} = \frac{N_{i,\text{test}}^{(1)}}{T - S_i}, \qquad \widehat{\mu}_i^{(0)} = \frac{N_{i,\text{test}}^{(0)}}{V_{i,\text{test}}^{(0)}}, \qquad \widehat{\mu}_i^{(1)} = \frac{N_{i,\text{test}}^{(1)}}{V_{i,\text{test}}^{(1)}}.$$

Under $\mathcal{H}_0^{(i)}$, the estimators are

$$\widehat{b}_i = \frac{N_{i,\text{test}}}{T}, \qquad \widehat{\mu}_i = \frac{N_{i,\text{test}}}{V_{i,\text{test}}}.$$

The likelihood-ratio statistic is then defined by

$$D_i^{\text{test}} = 2\left[\ell_i^{\text{test}}\left(S_i; \widehat{b}_i^{(0)}, \widehat{b}_i^{(1)}, \widehat{\mu}_i^{(0)}, \widehat{\mu}_i^{(1)}\right) - \ell_i^{\text{test}}\left(T; \widehat{b}_i, \widehat{b}_i, \widehat{\mu}_i, \widehat{\mu}_i\right)\right]. \tag{3.3}$$

We now study the asymptotic behaviour of the test statistic under the null hypothesis. We consider a large-time regime where $T \to \infty$. In this regime, the candidate change-point S_i^{cand} is allowed to depend on T, while the true model parameters remain fixed. The following proposition shows that, under this asymptotic regime, the likelihood-ratio statistic computed on the independent test sample has a standard chi-squared limit under the null hypothesis.

Proposition 1. *Let X_i be a marked Poisson process observed on $[0, T]$, split by the thinning method as in Equation (3.2). Let S_i^{cand} be the change point estimated in the selection sample which is allowed to depend on T. Assume that, as $T \to \infty$, $\frac{S_i^{\text{cand}}}{T} \longrightarrow s_i, \quad s_i \in (0, 1)$ and under the null hypothesis $\mathcal{H}_0^{(i)}$, the test sample is generated by a marked Poisson process with fixed parameters $b^\star$ and $\mu^\star$. Then the likelihood-ratio statistic D_i^{test} in Equation (3.3) converges in distribution to a chi-squared distribution with two degrees of freedom*

$$D_i^{\text{test}} \xrightarrow[T\to\infty]{d} \chi_2^2.$$

The proof is based on the standard likelihood-ratio theorem and is given in Appendix A.

Therefore, the p-value associated with process i is computed as

$$p_i = 1 - F_{\chi_2^2}\left(D_i^{\text{test}}\right), \qquad i = 1, \ldots, n.$$

Multiple testing correction.

We also apply the Benjamini–Hochberg procedure to the resulting p-values. The final estimated change-point is then defined as

$$S_i^{(M_{\max})} = \begin{cases} S_i^{\text{cand}}, & \text{if } \mathcal{H}_0^{(i)} \text{ is rejected,} \\ T, & \text{otherwise.} \end{cases}$$

Final parameter estimation.

After the testing step, we keep the final estimated change-points $S_i^{(M_{\max})}$. The original intensities are recovered from the selection-sample intensities as $\widehat{\lambda}^{(0)} = \frac{\widehat{a}^{(0)}}{q}, \qquad \widehat{\lambda}^{(1)} = \frac{\widehat{a}^{(1)}}{q}$. The final mark parameters are $\widehat{\mu}^{(0)}$ and $\widehat{\mu}^{(1)}$, estimated on the selection sample.

The complete procedure then

Algorithm 2 Offline data-splitting detection via thinning

Input: $X_1, \ldots, X_n$, T, $q \in (0,1)$, α, and iterations M_{sel}.
Output: Final change-points $S_1, \ldots, S_n$ and estimated parameters.
Step 1: Data Splitting via Thinning
for each process $i = 1, \ldots, n$ and each event j **do**
    Generate $B_{ij} \sim \text{Bernoulli}(q)$.
    if $B_{ij} = 1$ **then**: $(T_{ij}, U_{ij}) \in \mathcal{D}_i^{\text{sel}}$.
    else: $(T_{ij}, U_{ij}) \in \mathcal{D}_i^{\text{test}}$.
    end if
end for
Step 2: Selection Step (on selection sample)
Initialize $a^{(0)}, a^{(1)}, \mu^{(0)}, \mu^{(1)}$.
for $m = 1, \ldots, M_{\text{sel}}$ **do Compute**:

$$\widehat{S}_i^{(m)} = \arg\max_{S \in \mathcal{G}_i} \ell_i^{\text{sel}} \left(S; a^{(m-1)}, \mu^{(m-1)} \right).$$

Using $\widehat{S}_1^{(m)}, \ldots, \widehat{S}_n^{(m)}$, update $a^{(0,m)}, a^{(1,m)}, \mu^{(0,m)}, \mu^{(1,m)}$.
end for
Set $S_i^{\text{cand}} = \widehat{S}_i^{(M_{\text{sel}})}$ for all $i = 1, \ldots, n$.
Step 3: Testing Step (on independent test sample)
for $i = 1, \ldots, n$ **do**
    Compute:

$$D_i^{\text{test}} = 2 \left[\ell_i^{\text{test}} \left(S_i; \widehat{b}_i^{(0)}, \widehat{b}_i^{(1)}, \widehat{\mu}_i^{(0)}, \widehat{\mu}_i^{(1)} \right) - \ell_i^{\text{test}} \left(T; \widehat{b}_i, \widehat{b}_i, \widehat{\mu}_i, \widehat{\mu}_i \right) \right].$$

    Compute the p-value $p_i = 1 - F_{\chi_2^2}(D_i^{\text{test}})$, apply the Benjamini–Hochberg procedure to $p_1, \ldots, p_n$.
    Set $S_i = S_i^{\text{cand}}$ if $\mathcal{H}_0^{(i)}$ is rejected; otherwise set $S_i = T$.
    return $S_1, \ldots, S_n$ and test sample estimators.

Remark.

The selection–testing procedure relies on a thinning step, which reduces the amount of data available at each stage.

This limitation is important for small datasets. In the testing step, the parameters $b_i^{(0)}$, $b_i^{(1)}$, $\mu_i^{(0)}$ and $\mu_i^{(1)}$ are re-estimated using only the events of process i in the test sample. Therefore, when process i contains only a small number of events, the test sample may not contain enough information to obtain stable parameter estimates or a reliable likelihood-ratio statistic. In such cases, the asymptotic χ^2_2 approximation may be inaccurate and the procedure may lose power.

For this reason, Algorithm 2 is less suitable for the small dataset considered in this internship, as shown in Chapter 5. The procedure is therefore mainly presented as a reference method for larger datasets, longer observation horizons, or future extensions where more events are available.

Chapter 4

Online Change-Point Detection Algorithm

In the offline setting, the complete trajectory of the marked Poisson process is observed on a fixed time interval $[0, T]$ before any statistical decision is made. In contrast, in the online setting, observations are received sequentially over time. More precisely, we choose a monitoring step $\Delta > 0$ and observe the process at times $t_d = d\Delta, \quad d = 1, 2, \ldots$. After each new observation period $[t_{d-1}, t_d)$, the procedure updates the information available up to time t_d and decides whether an alarm should be raised. In this framework, raising an alarm A means that the procedure detects a change in the behaviour of the process. Moreover, we know that detecting a change-point implies that the process is under pre-change state for a certain duration before any change-point occurs. So, it is naturally assumed that a change-point will not happen during the very first few monitoring steps. To account for this initial period, we introduce a minimum monitoring time index $d_{\min} \geq 0$. Afterwards, the testing procedure effectively begins, and alarms are allowed to be raised, only starting from time $t_{d_{\min}}$. The objective is therefore to detect the change as soon as possible after it occurs, while controlling the probability of false alarms [4].

Since the procedure is applied repeatedly over several processes and several monitoring times, the relevant error is a global false alarm. In this chapter, we use a family-wise error rate (FWER) control: under the global null hypothesis that there are no change-points across all processes, we aim to control the probability of raising at least one false alarm over the entire collection of processes and over the whole monitoring period [11]. Furthermore, within the online framework considered in this report, we assume that the true parameters λ and μ are known in advance and are provided by the offline algorithm.

4.1 Parametric Bootstrap Algorithm with Ram Effect

We first consider a parametric bootstrap online procedure for monitoring n independent marked Poisson processes over a finite horizon $[0, T]$, inspired by [4]. In this version, we include the ram effect, as in Algorithm 1. For each process i, the observations are written as $X_i = \{(T_{ij}, U_{ij}, R_{ij}) : j = 1, \ldots, N_i\}$. At each monitoring time t_d, the procedure tests whether a change-point has occurred somewhere in the past interval $[0, t_d)$. Thus, it uses the whole trajectory observed up to t_d, not only the observations collected during the last monitoring interval.

Observed statistic.

For each process $i \in \{1, \ldots, n\}$ and each monitoring time t_d, we test

$$\mathcal{H}_{0,i}^d : \text{ no change-point has occurred for process } i \text{ on } [0, t_d),$$

against

$$\mathcal{H}_{1,i}^d : \text{ there exists a change-point } S_i \in [0, t_d).$$

Under $\mathcal{H}_{0,i}^d$, the log-likelihood contribution of process i is

$$\ell_{0,i}^d(\theta) = \sum_{r=1}^{R} \left[N_{ir}([0, t_d)) \log\left(\lambda_r^{(0)} \mu_r^{(0)}\right) - \lambda_r^{(0)} t_d - \mu_r^{(0)} V_{ir}([0, t_d)) \right],$$

where $\theta = \{\lambda_r^{(0)}, \lambda_r^{(1)}, \mu_r^{(0)}, \mu_r^{(1)} : r = 1, \ldots, R\}$.

Under $\mathcal{H}_{1,i}^d$, for a candidate change-point $S \in [0, t_d)$, the log-likelihood is

$$\ell_{1,i}^d(S; \theta) = \sum_{r=1}^{R} \Big[N_{ir}([0, S]) \log\left(\lambda_r^{(0)} \mu_r^{(0)}\right) - \lambda_r^{(0)} S - \mu_r^{(0)} V_{ir}([0, S]) + N_{ir}((S, t_d)) \log\left(\lambda_r^{(1)} \mu_r^{(1)}\right) - \lambda_r^{(1)} (t_d - S) - \mu_r^{(1)} V_{ir}((S, t_d)) \Big].$$

The estimated change-point at time t_d is $\widehat{S}_{i,d} \in \operatorname{argmax}_{S \in [0, t_d)} \ell_{1,i}^d(S; \theta)$. The generalized likelihood-ratio statistic is

$$G_i(d) = \ell_{1,i}^d(\widehat{S}_{i,d}; \theta) - \ell_{0,i}^d(\theta).$$

Bootstrap calibration.

As in Algorithm 1, the null distribution of the statistic is approximated by parametric bootstrap. The difference is that the calibration must now

account for the repeated testing over all processes and all monitoring times. In the present version, the bootstrap simulation also includes the ram effect.

Under the global null hypothesis

$$\mathcal{H}_0 : \text{ no process has a change-point during } [0, T],$$

we generate M_{boot} bootstrap samples. For each bootstrap replication $b = 1, \ldots, M_{\text{boot}}$ and each process $i = 1, \ldots, n$, we simulate one null trajectory by generating independent marked Poisson processes for each ram $r = 1, \ldots, R$, with intensity $\lambda_r^{(0)}$ and mark rate $\mu_r^{(0)}$ provided by the offline algorithm. The simulated events from all rams are then pooled together and ordered by time, while keeping their ram labels. This gives one bootstrap null trajectory for process i.

For each simulated trajectory and each monitoring time t_d, we compute the same statistic

$$G_{b,i}^0(d) = \ell_{1,b,i}^d(\widehat{S}_{b,i,d}^0; \theta) - \ell_{0,b,i}^d(\theta),$$

where

$$\widehat{S}_{b,i,d}^0 \in \underset{S \in [0, t_d)}{\arg\max}\, \ell_{1,b,i}^d(S; \theta).$$

The bootstrap p-value of process i at time t_d is then

$$\widehat{p}_i(d) = \frac{1}{M_{\text{boot}} n} \sum_{b=1}^{M_{\text{boot}}} \sum_{j=1}^{n} \mathbf{1}\left\{G_{b,j}^0(d) \geq G_i(d)\right\}.$$

Global threshold.

To control the probability of at least one false alarm, we calibrate the threshold from the minimum bootstrap p-value over all processes and all monitoring times. For each bootstrap replication b, define

$$\widehat{p}_{b,i}^0(d) = \frac{1}{M_{\text{boot}} n} \sum_{b'=1}^{M_{\text{boot}}} \sum_{j=1}^{n} \mathbf{1}\left\{G_{b',j}^0(d) \geq G_{b,i}^0(d)\right\},$$

and

$$M_b^0 = \min_{1 \leq i \leq n} \min_{d_{\min} \leq d \leq D} \widehat{p}_{b,i}^0(d),$$

where $d_{\min}$ is the first monitoring index from which alarms are allowed. Then, the global alarm threshold is chosen as

$$\widehat{u}_\alpha = \sup\left\{u \in [0,1] : \frac{1}{M_{\text{boot}}} \sum_{b=1}^{M_{\text{boot}}} \mathbf{1}_{\{M_b^0 \leq u\}} \leq \alpha\right\}.$$

Online monitoring rule.

Let $\mathcal{I}_1 = \{1, \ldots, n\}$ be the set of active processes at the beginning of the monitoring period. At each monitoring time t_d, we compute $\widehat{p}_i(d)$ for all active processes $i \in \mathcal{I}_d$. If $d \geq d_{\min}$, an alarm is raised for process i whenever $\widehat{p}_i(d) \leq \widehat{u}_\alpha$. The set of detected processes at time t_d is

$$\mathcal{J}_d = \{i \in \mathcal{I}_d : \widehat{p}_i(d) \leq \widehat{u}_\alpha\}.$$

For each $i \in \mathcal{J}_d$, the alarm time is recorded as $A_i = t_d$, and the estimated change-point is recorded as $\widehat{S}_i = \widehat{S}_{i,d}$. Once a process has raised an alarm, $\mathcal{I}_{d+1} = \mathcal{I}_d \setminus \mathcal{J}_d$. If a process never raises an alarm before the final monitoring time, we set $A_i = +\infty$.

The complete procedure is then

Algorithm 3 Online parametric bootstrap detection with ram effect

Input: $X_1, \ldots, X_n$, T, Δ, Δ, θ,$\mathrm{M}_{\mathrm{boot}}$, α, and $d_{\min}$.
Output: Alarm times A_i, estimated change-points $\widehat{S}_i$.
Bootstrap: Simulate M_{boot} null datasets.
for $b = 1, \ldots, M_{\mathrm{boot}}$, $i = 1, \ldots, n$, and $d = 1, \ldots, D$ **do**
    Compute $G^0_{b,i}(d)$.
end for
for $b = 1, \ldots, M_{\mathrm{boot}}$, $i = 1, \ldots, n$, and $d = 1, \ldots, D$ **do**
    Compute $\widehat{p}^0_{b,i}(d)$.
end for
for $b = 1, \ldots, M_{\mathrm{boot}}$ **do** Compute M^0_b.
end for
Choose $\widehat{u}_\alpha$ and initialize $\mathcal{I}_1 = \{1, \ldots, n\}$, $A_i = +\infty$, $\widehat{S}_i = \mathrm{NA}$.
for $d = 1, \ldots, D$ **do**
    for each active process $i \in \mathcal{I}_d$ **do** Compute $G_i(d)$, $\widehat{S}_{i,d}$, and $\widehat{p}_i(d)$.
    end for
    if $d \geq d_{\min}$ **then** Set $\mathcal{J}_d = \{i \in \mathcal{I}_d : \widehat{p}_i(d) \leq \widehat{u}_\alpha\}$.
        for each $i \in \mathcal{J}_d$ **do** Set $A_i = t_d$ and $\widehat{S}_i = \widehat{S}_{i,d}$.
        end for
        Update $\mathcal{I}_{d+1} = \mathcal{I}_d \setminus \mathcal{J}_d$.
    else Set $\mathcal{I}_{d+1} = \mathcal{I}_d$.
    end if
end for
Return A_i and $\widehat{S}_i$. =0

4.2 E-valued Method for an Infinite Monitoring Horizon

The bootstrap procedure described in Section 4.1 is designed for a finite monitoring horizon. Indeed, the threshold is calibrated by simulating complete null trajectories on a fixed interval $[0, T]$. This construction is therefore tied to a pre-specified terminal time T. If the monitoring horizon is extended, or if new monitoring times are added after the calibration step, the previously computed threshold no longer automatically guarantees the desired global type I error control. [17]

This limitation is important in online change-point detection. In many practical situations, monitoring is not naturally stopped at a deterministic final time. Data may arrive continuously, and the analyst may want to raise an alarm as soon as there is convincing evidence of a change, without deciding in advance how long the monitoring period will last. In such an open-ended setting, a direct bootstrap calibration over all future monitoring times is not feasible. To address this issue, we use the framework of e-values and e-processes [15]. An e-value is a nonnegative statistic whose expectation under the null hypothesis is at most one. Large e-values indicate evidence against the null. An e-process is a nonnegative sequential process $(Y_d)_{d\geq 0}$ such that, for any stopping time τ with respect to the filtration $(\mathcal{F}_d)_{d\geq 0}$, the stopped value Y_τ remains an e-value. This property makes e-processes suitable for online monitoring over an unknown or infinite horizon.

In the present Poisson monitoring problem, we construct the e-process by multiplying blockwise likelihood ratios between the post-change and pre-change models. Under the no-change null hypothesis

$$\mathcal{H}_0 : \text{ no change-point occurs in any process,}$$

we will prove that this product is a nonnegative martingale and therefore can be monitored sequentially. An alarm is raised when the process exceeds $1/\alpha$. Compared with the bootstrap method, this approach is less sensitive because it does not scan all past change-point locations, but it remains valid over an unknown or infinite monitoring horizon.

Monitoring blocks and local hypotheses.

At each monitoring time t_d, we consider the new block of observations collected during the interval $[t_{d-1}, t_d)$:

$$X_d = X([t_{d-1}, t_d)) = \{(T_j, U_j) : t_{d-1} \leq T_j < t_d\}.$$

We denote by $N_d = N([t_{d-1}, t_d))$ the number of events in this block, and by $V_d = V([t_{d-1}, t_d))$ the corresponding sum of marks. The information available up to time t_d is represented by the filtration $\mathcal{F}_d = \sigma(X_1, \ldots, X_d)$.

At each monitoring time t_d, we compare the pre-change model with the post-change model on the most recent block. The local null hypothesis is

$$\mathcal{H}_{0,d} : X_d \text{ is generated by the pre-change model } (\lambda^{(0)}, \mu^{(0)}),$$

against

$$\mathcal{H}_{1,d} : X_d \text{ is generated by the post-change model } (\lambda^{(1)}, \mu^{(1)}).$$

Equivalently, $\mathcal{H}_{1,d}$ means that the new monitoring block $[t_{d-1}, t_d)$ is tested as being generated by the post-change regime.

Block likelihood ratio.

Let f_0 and f_1 denote the probability densities of one monitoring block under the pre-change model $(\lambda^{(0)}, \mu^{(0)})$ and the post-change model $(\lambda^{(1)}, \mu^{(1)})$, respectively. Under $\mathcal{H}_{0,d}$, the log-likelihood of the block X_d is

$$\ell_{0,d} = N_d \log(\lambda^{(0)}\mu^{(0)}) - \lambda^{(0)}\Delta - \mu^{(0)}V_d.$$

Under $\mathcal{H}_{1,d}$, the corresponding log-likelihood is

$$\ell_{1,d} = N_d \log(\lambda^{(1)}\mu^{(1)}) - \lambda^{(1)}\Delta - \mu^{(1)}V_d.$$

We define the block likelihood ratio by

$$L_d = \frac{f_1(X_d)}{f_0(X_d)} = \exp(\ell_{1,d} - \ell_{0,d}).$$

Thus,

$$L_d = \exp\big\{-(\lambda^{(1)} - \lambda^{(0)})\Delta\big\} \left(\frac{\lambda^{(1)}\mu^{(1)}}{\lambda^{(0)}\mu^{(0)}}\right)^{N_d} \exp\big\{-(\mu^{(1)} - \mu^{(0)})V_d\big\}.$$

We then define the process $(Y_d)_{d\geq 0}$ such that

$$Y_0 = 1, \qquad \text{and} \qquad Y_d = \prod_{k=1}^{d} L_k \quad \text{for } d \geq 1. \tag{4.1}$$

The alarm time A is defined as the first time at which the e-process crosses the threshold $1/\alpha$

$$A = \inf\left\{t_d : \ d \geq 1, \ Y_d \geq \frac{1}{\alpha}\right\}.$$

The following proposition establishes that this sequential monitoring procedure successfully controls the family-wise error rate (FWER) under the global null hypothesis that no change-point occurs in any process.

Proposition 2. *Let $\alpha \in (0,1)$, and let $X_d = X([t_{d-1}, t_d))$ be the monitoring block observed at step d, with filtration $\mathcal{F}_d = \sigma(X_1, \ldots, X_d)$. Let f_0 and f_1 be the densities of one monitoring block under the pre-change and post-change models, respectively. Assume that, under the global null hypothesis, each block X_d has density f_0 and is independent of $\mathcal{F}_{d-1}$. Define the process $(Y_d)_{d\geq 0}$ as in Equation 4.1. Let*

$$A = \inf\left\{t_d : \ d \geq 1, \ Y_d \geq \frac{1}{\alpha}\right\},$$

with $A = +\infty$ if the set is empty. Then

$$\mathbb{P}_{\mathcal{H}_0}(A < +\infty) \leq \alpha.$$

The proof of Proposition 2 is given in Appendix B.
By Proposition 2, since FWER $= \mathbb{P}_{\mathcal{H}_0}((A < +\infty) \leq \alpha$, and under the global null hypothesis, each block X_d is independent of the past $\mathcal{F}_{d-1}$, the proposed monitoring procedure effectively controls the infinite-horizon family-wise error rate at level α.

Remark.

The construction presented above is written for one single ewe, or equivalently for one marked Poisson process. In principle, the same e-process approach can also be extended to several independent processes by multiplying the block likelihood ratios across processes. In that case, the resulting alarm is a global alarm: it indicates that at least one process may have a change-point, but it does not identify which process is responsible for the change.

Another possibility is to construct one e-process for each process and use a multiple-testing correction. For example, in this internship, we use the Bonferroni method [11], with a threshold of α/n for each process. This would control the probability of making at least one false detection among all processes. However, this approach is usually conservative and may lose power when the number of processes is large. Furthermore, a major limitation of this method compared to the bootstrap procedure is its inability to estimate the exact change-point S_i. The e-process only determines the alarm time A_i when the threshold is crossed, but it does not provide an estimator $\widehat{S}_i$ for the structural break. For these reasons, the e-process method is mainly presented here as a reference procedure for situations with a very long, unknown, or infinite monitoring horizon. In our data setting, it is less suitable than the finite-horizon bootstrap procedure described in Section 4.1, which uses the observed trajectory more efficiently and is better adapted to our practical monitoring problem.

The complete algorithm of this procedure is

Algorithm 4 Online e-process monitoring with Bonferroni correction

Input: $X_1, \ldots, X_n$, D, Δ, $\theta = (\lambda^{(0)}, \lambda^{(1)}, \mu^{(0)}, \mu^{(1)})$, and α.
Output: Alarm times A_i for each process i.
Initialization:
    Set Bonferroni threshold $\tau = \frac{n}{\alpha}$.
    Initialize active set $\mathcal{I}_1 = \{1, \ldots, n\}$.
    Set $A_i = +\infty$ and $Y_{i,0} = 1$ for all $i = 1, \ldots, n$.
for $d = 1, \ldots, D$ **do**
    for each $i \in \mathcal{I}_d$ **do**
        Compute

$$L_{i,d} = \exp\{-(\lambda^{(1)} - \lambda^{(0)})\Delta\} \left(\frac{\lambda^{(1)}\mu^{(1)}}{\lambda^{(0)}\mu^{(0)}}\right)^{N_{i,d}} \exp\{-(\mu^{(1)} - \mu^{(0)})V_{i,d}\}$$

        Update the e-process: $Y_{i,d} = Y_{i,d-1}L_{i,d}$.
    end for
    $\mathcal{J}_d = \{i \in \mathcal{I}_d : Y_{i,d} \geq \tau\}$.
    for each $i \in \mathcal{J}_d$ **do**
        Set $A_i = t_d$.
    end for
    Update active processes: $\mathcal{I}_{d+1} = \mathcal{I}_d \setminus \mathcal{J}_d$.
    if $\mathcal{I}_{d+1} = \emptyset$ **then**
        break
    end if
end for
Return: A_i for $i = 1, \ldots, n$.

Chapter 5

Case Study on Experimental Data

In this chapter, we will apply all the algorithms 1, 2, 3, and 4 from Chapters 3 and 4 to both the simulated data and the dataset presented in Chapter 2.

5.1 Validation on simulated data

5.1.1 Validation without ram effect

We first validate the algorithms in a simplified setting without ram effect. In this setting, all processes are simulated with the same pre-change and post-change parameters. To evaluate the algorithms more reliably, we apply them to several independently simulated datasets. In each replication, we generate N independent marked Poisson processes with parameters $\lambda_0, \lambda_1, \mu_0, \mu_1$ on $[0, T]$. The number of null processes, corresponding to trajectories with no change-point and $S_i = T$, is sampled uniformly. For the remaining processes, the true change-points S_i are generated independently from a uniform distribution $\mathcal{U}(t_{\min}, t_{\max})$, with $0 < t_{\min} < t_{\max} < T$. For the online algorithms, the parameters $\lambda_0, \lambda_1, \mu_0, \mu_1$ are used directly in the monitoring rules. Tables 5.1 and 5.2 give an illustrative summary of the results for the offline and online methods, respectively. The results show different behaviours across the four methods.

- **Offline bootstrap method.** As shown in Table 5.1, Algorithm 1 performs well: it detects most true change-points, identifies most null processes, and gives small localization errors.
- **Selection-testing method.** Algorithm 2 is more conservative. It misses more true change-points, especially when the rupture is close to

the beginning or the end of the observation window. It's because that after thinning, one of the two regimes may contain too few events for the test to have enough power.

- **Online bootstrap method.** As shown in Table 5.2, Algorithm 3 detects most true changes, keeps false alarms low, and raises alarms with a reasonable delay.
- **E-process method.** Algorithm 4 is more conservative. It misses more true change-points and raises alarms later.

Method	TP	FP	TN	FN	Mean absolute error of $\widehat{S}_i$	Variance of error
Offline bootstrap	1212	27	1258	3	1.127	3.260
Selection-testing	811	6	1279	404	10.258	223.251

Table 5.1: Application of Algorithms 1 and 2 under the simulation setup described in Section 5.1.1. Results are computed over 50 independent simulated datasets with $T = 50$, $N = 50$, $d_{\min} = 1$, $\lambda_0 = 1$, $\lambda_1 = 3$, $\mu_0 = 2$, and $\mu_1 = 1$. See Appendix C for notation and conventions.

Method	TP	FP	TN	FN	Mean absolute error of $\widehat{S}_i$	Variance of error	Mean alarm delay
E-process	818	2	1283	397	NA	NA	12.261
Online bootstrap	1201	5	1280	14	1.202	5.005	4.134

Table 5.2: Application of Algorithms 3 and 4 under the simulation setup described in Section 5.1.1. Results are computed over 50 independent simulated datasets with $T = 50$, $N = 50$, $d_{\min} = 1$, $q = 0.5$, $\lambda_0 = 1$, $\lambda_1 = 3$, $\mu_0 = 2$, and $\mu_1 = 1$. See Appendix C for notation and conventions.

5.1.2 Validation with ram effect

We now check whether adding a ram-effect parameter improves the detection results when different rams have different activity patterns. To do this, we simulate data with several rams and compare two versions of Algorithm 1. In the first version, all ram labels are collapsed into a single group, so the algorithm is applied as if there were no ram effect. In the second version, the

ram labels are kept, and separate pre-change and post-change parameters are estimated for each ram. We also apply the online procedure in Algorithm 3 to the same simulated data, in order to check whether the online method can recover the same change-points with a reasonable alarm delay.

The purpose of this simulation is to isolate the effect of ram heterogeneity. If the rams have different activity levels before and after the change-point, collapsing them into a single group may hide the real change in behaviour. In contrast, the ram-effect model can distinguish the contribution of each ram and should therefore recover the change-points more accurately.

The difference between the collapsed model and the ram-specific model can be seen in the illustrative results reported in Tables 5.3 and 5.4.

Ewe	S_{true}	$\widehat{S}_{\text{offline,one ram}}$	$\widehat{S}_{\text{offline,several rams}}$	$\widehat{S}_{\text{online,several rams}}$	$A_{\text{online,several rams}}$
1	15	80.00	15.41	15.04	17
2	20	80.00	20.07	19.25	21
3	25	80.00	25.27	24.94	26
4	30	80.00	30.06	29.66	31
5	40	78.02	40.22	39.95	41
6	50	80.00	50.30	49.85	52
7	60	80.00	60.05	59.98	63
8	80	80.00	80.00	80.00	80
9	80	80.00	80.00	80.00	80
10	80	80.00	80.00	80.00	80

Table 5.3: Estimated change-points and online alarm times obtained using Algorithm 1 with collapsed ram labels, Algorithm 1 with ram-specific parameters, and Algorithm 3 with ram-specific parameters. The simulation uses $N = 10$, $R = 2$, $T = 80$, $d_{\min} = 1$, and $\alpha = 0.05$.

Case	Ram	$\lambda^{(0)}_{\text{true}}$	$\widehat{\lambda}^{(0)}$	$\lambda^{(1)}_{\text{true}}$	$\widehat{\lambda}^{(1)}$	$\mu^{(0)}_{\text{true}}$	$\widehat{\mu}^{(0)}$	$\mu^{(1)}_{\text{true}}$	$\widehat{\mu}^{(1)}$
Collapsed rams	all	4.50	4.519	4.50	1.008	2.00	2.151	2.00	1.733
Ram-specific model	ram1	4.00	4.018	0.50	0.521	2.00	2.009	1.00	0.936
Ram-specific model	ram2	0.50	0.451	4.00	4.052	2.00	2.118	3.00	2.960

Table 5.4: True and estimated pre-change and post-change parameters for the collapsed one-ram model and the ram-specific model. The table reports $\lambda^{(0)}$, $\lambda^{(1)}$, $\mu^{(0)}$, and $\mu^{(1)}$ under the same simulation setup as Table 5.3, with $N = 10$, $R = 2$, $T = 80$, $d_{\min} = 1$, and $\alpha = 0.05$.

5.2 Application of the online and offline algorithms to the datasets

In this section, we apply Algorithms 1 and 3 to our datasets. These procedures are uniquely suited to our data, which feature short observation periods, limited sample sizes, and non-uniform event distributions across rams (Section 2.2). By using a parametric bootstrap framework, they ensure reliable error control without relying on large-sample asymptotics. Moreover, by capturing ram-specific intensities, they can account for differences in ram activity. We test them first on the epic2014 dataset, and then on six other datasets: La Fage (2023, 2025), La Cazotte (2023, 2024), and Corse (2024, 2025).

As noted in Section 2.2, the recorded interaction durations are not fully reliable, so we do not use them in our analysis. In practice, the mark parameters are fixed before and after the change-point, so we set $\mu_r^{(0)} = \mu_r^{(1)} = 1$ for all $r = 1, \ldots, R$. For the online algorithm, we then use the ram-specific intensities $\lambda_r^{(0)}$ and $\lambda_r^{(1)}$ obtained from the offline step. We then compare the results of the offline and online algorithms, and we also compare the obtained rupture detections with the manual detection rule provided by Nathalie Debus.

Additionally, as noted in Section 2.2, two specific rams, with IDs 11171 and 93202, appear in multiple datasets. We then use them to check whether the intensity of the same ram stays consistent across different experiments. Finally, our datasets do not all cover the same number of observation days. Therefore, the time scale used for the online alarm rule is adapted to the length of each experiment. For the shorter datasets, namely Corse 2024, Corse 2025, and La Fage 2023, the alarm condition is checked every hour. For the remaining datasets, the alarm condition is checked once per day.

Comparison of offline and online change-point estimators.

The offline and online procedures give similar results on most datasets, as summarized in Table 5.5. In particular, the number of coincident change-point estimates is generally larger than the number of different estimates. The mean alarm delay also remains reasonable, which indicates that the online procedure usually raises an alarm shortly after the estimated change-point.

Figure 5.1 illustrates this behaviour on La Fage 2025: many offline and online estimates coincide, and the alarm times remain close to the detected changes.

Dataset	Same S	Different S	Mean alarm delay
EPIC 2014	132	21	0.427 days
Corse 2024	23	8	0.944 hours
Corse 2025	23	3	7.432 hours
La Fage 2023	23	9	2.164 hours
La Fage 2025	38	6	0.525 days
La Cazotte 2023	63	15	2.076 days
La Cazotte 2024	68	15	0.748 days

Table 5.5: Summary of the agreement between Algorithms 1 and 3 across datasets, with $\mu_r^{(0)} = \mu_r^{(1)} = 1$, $n_{\text{prec}} = 50$, $M_{\text{boot}} = 500$, and $\alpha = 0.05$. Algorithm 3 uses λ estimated by Algorithm 1.

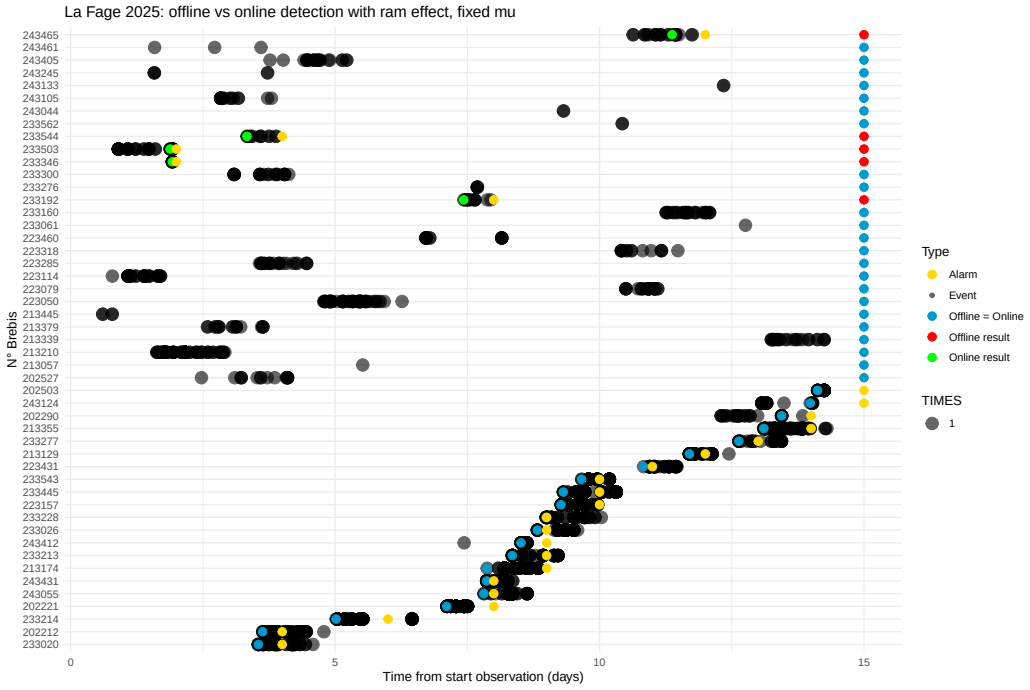


Figure 5.1: Algorithms 1 and 3 detection results for La Fage 2025 with $\mu_r^{(0)} = \mu_r^{(1)} = 1$, $M_{\text{boot}} = 500$, $\alpha = 0.05$, and $d_{\min} = 1$. Black points are observed events, red points are offline estimates, green points are online estimates, yellow points are online alarms, and blue points indicate coincident offline and online estimates.

Comparison with non-automated detection.

The event-level comparison in Table 5.6 is based on an empirical classification method provided by Nathalie Debus, as mentioned in Chapter 2.

This method compares the offline rupture decision with a non-automated rule based on the cumulative number of Alpha-Detector readings within each day. Most of these event records are classified as `OKOK`, while only a small fraction are classified as `KOOK`. In contrast, when the manual rule gives a negative signal, the offline decisions are more balanced between `OKKO` and `KOKO`. This suggests that the offline algorithm agrees well with strong manual signals, but may also detect ruptures before the manual daily-count rule becomes strong. The categories `OK?` and `KO?` remain uncertain manual cases.

Dataset	`OKKO`	`OK?`	`OKOK`	`KOKO`	`KO?`	`KOOK`
EPIC 2014	248	366	3128	48	47	80
Corse 2024	28	42	189	28	20	5
Corse 2025	26	39	107	21	14	0
La Fage 2023	22	33	232	38	40	25
La Fage 2025	38	57	1095	40	15	30
La Cazotte 2023	60	90	1115	84	65	106
La Cazotte 2024	32	48	328	116	90	61

Table 5.6: Event-level comparison between the offline rupture detection algorithm and the empirical manual classification rule. The notation and conventions are given in Appendix C.

Comparison of repeated ram effects across years.
To check whether the same ram has a stable effect across different experiments, we compare the estimated intensities for two ram IDs observed in two different years within the same site. The results in Table 5.7 show that the estimated intensities are not stable across years. This suggests that we cannot assume that the effect of the same ram remains identical across different experiments or years. These variations may also reflect factors other than the individual ram itself, such as age, breed, or year-specific experimental conditions. This point could be investigated further in the continuation of the internship.

Ram ID	Site	$\widehat{\lambda}^{(0)}_{\text{Year 1}}$	$\widehat{\lambda}^{(1)}_{\text{Year 1}}$	$\widehat{\lambda}^{(0)}_{\text{Year 2}}$	$\widehat{\lambda}^{(1)}_{\text{Year 2}}$
11171	La Cazotte	0.022	0.086	0.058	0.097
93202	Corse	0.024	0.367	0.001	0.001

Table 5.7: Comparison of estimated ram-specific intensities for repeated ram IDs across years of Ram ID 11171 and ram ID 93202.

Chapter 6

Conclusion and Perspectives

This internship focused on change-point detection methods for marked Poisson processes, with an application to automated oestrus detection in small ruminants. The main objective was to detect, for each ewe, a change in the rhythm of mounting events that could be interpreted as a change in reproductive behaviour. Both offline and online detection settings were considered.

First, we studied an existing likelihood-based offline algorithm and adapted it to include a ram effect. This extension was important because the experimental data showed that the number of events was not uniformly distributed across rams. Allowing each ram to have its own pre-change and post-change intensities made the model more suitable for the available datasets. The simulation study confirmed this point: when the ram effect was ignored, the algorithm could fail to recover the true change-points, whereas the ram-effect version gave more accurate estimates.

Second, we studied an online parametric bootstrap procedure with ram effect, using the parameters estimated by the offline algorithm. This method is closer to the practical objective of the SynchroBIO project, since it raises alarms sequentially as new data arrive. On simulated data, the online algorithm detected the change-points with a moderate delay. On the experimental datasets, the comparison between offline and online results showed generally consistent behaviour. In several cases, the online estimated change-points were close to the offline estimates, and the alarm times occurred shortly after the online estimated change-points.

As discussed in Section 2.2, the recorded interaction durations were not considered fully reliable. For this reason, the main analysis was carried out without using the duration as a mark, by fixing $\mu_r^{(0)} = \mu_r^{(1)} = 1$ for all rams. In addition, the datasets had different observation lengths and different numbers of recorded events. The online monitoring grid therefore had to be adapted to the length of each experiment.

Another important observation concerns the stability of ram-specific intensities across years. Some ram IDs were observed in two different years within the same site, but their estimated intensities were not stable from one year to another. This suggests that the effect of the same ram cannot be assumed to remain identical across experiments. Consequently, the online algorithm should not rely on fixed ram-specific parameters estimated from a previous year. Instead, these parameters should be re-estimated for each dataset before applying the online procedure. These variations may also reflect factors other than the individual ram itself, such as age, breed, physiological condition, management context, or year-specific experimental conditions. This point could be investigated further in future work.

Several perspectives follow from this work. In the short term, an important next step is to compare the detected change-points with the artificial insemination information available in the data. This comparison would help assess whether the detected change-points are biologically meaningful and whether they occur at useful times for planning insemination. Such a validation step would provide a direct link between the statistical detections and the agronomic objective of the project.

A second perspective is to extend the model to the case of multiple change-points. In this report, each ewe is assumed to have at most one change-point. However, in practice, an animal's behaviour may change several times during the observation period, for example at the beginning and at the end of heat. During this internship, a two-change-point extension was considered, but it was not retained because no satisfactory testing procedure was found to distinguish reliably between zero, one, and two change-points. Multiple change-point methods have been studied in the literature in different contexts and methodological frameworks [7, 14], and these approaches could provide a useful basis for extending the present model to longer experiments or more complex behavioural patterns.

Finally, a natural direction for future work is to develop an online algorithm that does not require the pre-change and post-change intensities to be known in advance. The results on repeated ram IDs in Table 5.7 show that ram-specific intensities may vary across years and experimental contexts. This motivates adaptive online methods in which the parameters are updated as new data arrive. Such methods would be more robust for future multi-year experiments, where the ram effects are not known beforehand and cannot simply be transferred from one dataset to another.

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Appendix A

Proof of Proposition 1

In this section, we prove Proposition 1, which justifies the chi-squared convergence used in the testing step. To prove the convergence of the likelihood ratio statistic to a chi-square distribution under $\mathcal{H}_0$, we use the quadratic-form representation of the likelihood ratio statistic, as in [18]. This representation requires the asymptotic normality of the estimator, the convergence in probability of the normalized Hessian, and the identification of the null restriction as a two-dimensional smooth constraint. Unless stated otherwise, all the following arguments are made under $\mathcal{H}_0$.

Proof. Let us begin by observing how the two time intervals behave in the large-time asymptotic regime. We define $L_T^{(0)} = S_i^{\text{cand}}$, $L_T^{(1)} = T - S_i^{\text{cand}}$.

So

$$\frac{L_T^{(0)}}{T} \xrightarrow[T\to\infty]{} s_i, \qquad \frac{L_T^{(1)}}{T} \xrightarrow[T\to\infty]{} 1 - s_i.$$

In particular, both $L_T^{(0)}$ and $L_T^{(1)}$ tend to infinity as T tends to infinity. Under $\mathcal{H}_0^{(i)}$, there is no change-point. We denote the true parameters on the test sample by $b^\star = (1-q)\lambda^\star$, $\mu^\star$.

Then, under $\mathcal{H}_0^{(i)}$, the two counts are independent and satisfy

$$N_{i,\text{test}}^{(0)} \sim \text{Poisson}(b^\star L_T^{(0)}), \qquad N_{i,\text{test}}^{(1)} \sim \text{Poisson}(b^\star L_T^{(1)}).$$

Conditionally on the counts, the marks are independent and exponentially distributed with parameter $\mu^\star$.

Let $\vartheta = (b, \mu, \delta_b, \delta_\mu)^\top$, where $b^{(0)} = b$, $b^{(1)} = b + \delta_b$, $\mu^{(0)} = \mu$, $\mu^{(1)} = \mu + \delta_\mu$.

Under $\mathcal{H}_0^{(i)}$, the true parameter is $\vartheta^\star = (b^\star, \mu^\star, 0, 0)^\top$. The log-likelihood on the test sample is

$$\begin{aligned}\ell_{\text{test,T}}(\vartheta) = &- bL_T^{(0)} + N_{i,\text{test}}^{(0)} \log b - \mu V_{i,\text{test}}^{(0)} + N_{i,\text{test}}^{(0)} \log \mu - (b+\delta_b)L_T^{(1)} \\ &+ N_{i,\text{test}}^{(1)} \log(b+\delta_b) - (\mu+\delta_\mu)V_{i,\text{test}}^{(1)} + N_{i,\text{test}}^{(1)} \log(\mu+\delta_\mu).\end{aligned}$$

Step 1: consistency of the maximum likelihood estimator.

In the unrestricted model, the maximum likelihood estimators on the two intervals are

$$\widehat{b}^{(C)} = \frac{N_{i,\text{test}}^{(C)}}{L_T^{(C)}}, \quad \widehat{\mu}^{(C)} = \frac{N_{i,\text{test}}^{(C)}}{V_{i,\text{test}}^{(C)}}.$$

By the law of large numbers for Poisson variables, under $\mathcal{H}_0$,

$$\frac{N_{i,\text{test}}^{(C)}}{L_T^{(C)}} \xrightarrow[T\to\infty]{\mathbb{P}} b^\star.$$

Moreover, under $\mathcal{H}_0$, conditionally on the number of events, the marks are independent exponential random variables with parameter $\mu^\star$. Hence, for each $C \in \{0,1\}$,

$$\frac{V_{i,\text{test}}^{(C)}}{N_{i,\text{test}}^{(C)}} \xrightarrow[T\to\infty]{\mathbb{P}} \frac{1}{\mu^\star}.$$

Since also

$$\frac{N_{i,\text{test}}^{(C)}}{L_T^{(C)}} \xrightarrow[T\to\infty]{\mathbb{P}} b^\star, \qquad C \in \{0,1\},$$

we obtain

$$\frac{V_{i,\text{test}}^{(C)}}{L_T^{(C)}} = \frac{V_{i,\text{test}}^{(C)}}{N_{i,\text{test}}^{(C)}} \frac{N_{i,\text{test}}^{(C)}}{L_T^{(C)}} \xrightarrow[T\to\infty]{\mathbb{P}} \frac{b^\star}{\mu^\star}, \qquad C \in \{0,1\}.$$

Therefore, by the continuous mapping theorem,

$$\widehat{\mu}^{(0)} = \frac{N_{i,\text{test}}^{(0)}/L_T^{(0)}}{V_{i,\text{test}}^{(0)}/L_T^{(0)}} \xrightarrow[T\to\infty]{\mathbb{P}} \mu^\star, \qquad \widehat{\mu}^{(1)} = \frac{N_{i,\text{test}}^{(1)}/L_T^{(1)}}{V_{i,\text{test}}^{(1)}/L_T^{(1)}} \xrightarrow[T\to\infty]{\mathbb{P}} \mu^\star.$$

Thus, $\widehat{b}^{(0)} \xrightarrow[T\to\infty]{\mathbb{P}} b^\star$, $\widehat{b}^{(1)} \xrightarrow[T\to\infty]{\mathbb{P}} b^\star$, $\widehat{\mu}^{(0)} \xrightarrow[T\to\infty]{\mathbb{P}} \mu^\star$, $\widehat{\mu}^{(1)} \xrightarrow[T\to\infty]{\mathbb{P}} \mu^\star$.

Consequently,

$$\widehat{\vartheta}_T = \left(\widehat{b}^{(0)}, \widehat{\mu}^{(0)}, \widehat{b}^{(1)} - \widehat{b}^{(0)}, \widehat{\mu}^{(1)} - \widehat{\mu}^{(0)}\right)^\top \xrightarrow[T\to\infty]{\mathbb{P}} (b^\star, \mu^\star, 0, 0)^\top = \vartheta^\star.$$

Step 2: asymptotic normality of the normalized score.

Let us first compute the score contribution on a unit time interval. Let N be the number of events in an interval of length 1, $V = \sum_{j=1}^N U_j$ and $U_j \overset{i.i.d.}{\sim} \text{Exp}(\mu^\star)$.

Under $\mathcal{H}_0^{(i)}$, $N \sim \text{Poisson}(b^\star)$. We define $Y = \begin{pmatrix} -1 + \dfrac{N}{b^\star} \\ -V + \dfrac{N}{\mu^\star} \end{pmatrix}$.

Since $\mathbb{E}(N) = \text{Var}(N) = b^\star$ and $\mathbb{E}(U_j) = \mu^\star \text{Var}(U_j) = \frac{1}{\mu^\star}$, we have

$$\mathbb{E}(Y) = 0.$$

For the variance of the first component,

$$\text{Var}\left(-1 + \frac{N}{b^\star}\right) = \text{Var}\left(\frac{N}{b^\star}\right) = \frac{\text{Var}(N)}{(b^\star)^2} = \frac{1}{b^\star}.$$

By the law of total variance,

$$\text{Var}\left(-V + \frac{N}{\mu^\star}\right) = \mathbb{E}\left[\text{Var}\left(-V + \frac{N}{\mu^\star} \,\middle|\, N\right)\right] + \text{Var}\left[\mathbb{E}\left(-V + \frac{N}{\mu^\star} \,\middle|\, N\right)\right] = \frac{b^\star}{(\mu^\star)^2}.$$

By the law of total covariance,

$$\begin{aligned} \text{Cov}\left(-1 + \frac{N}{b^\star}, -V + \frac{N}{\mu^\star}\right) \quad &= \mathbb{E}\left[\text{Cov}\left(-1 + \frac{N}{b^\star}, -V + \frac{N}{\mu^\star} \,\middle|\, N\right)\right] \\ &\quad + \text{Cov}\left[\mathbb{E}\left(-1 + \frac{N}{b^\star} \,\middle|\, N\right), \mathbb{E}\left(-V + \frac{N}{\mu^\star} \,\middle|\, N\right)\right] = 0. \end{aligned}$$

Consequently, $\text{Var}(Y) = M^\star$, where $M^\star = \begin{pmatrix} \dfrac{1}{b^\star} & 0 \\ 0 & \dfrac{b^\star}{(\mu^\star)^2} \end{pmatrix}$.

Now define the score contributions on the two intervals. For $k = 1, \ldots, L_T^{(0)}$, let N_{0k} be the number of events in the k-th unit interval of $(0, S_i^{\text{cand}}]$, and let V_{0k} be the corresponding total mark. Let

$$Y_{0k} = \begin{pmatrix} -1 + \dfrac{N_{0k}}{b^\star} \\ -V_{0k} + \dfrac{N_{0k}}{\mu^\star} \end{pmatrix} \text{ and } A^{(0)} = \sum_{k=1}^{L_T^{(0)}} Y_{0k}.$$

Similarly, for interval $(S_i^{\text{cand}}, T]$, we define Y_{1k} in the same way and set $A^{(1)} = \sum_{k=1}^{L_T^{(1)}} Y_{1k}$.

The vectors $A^{(0)}$ and $A^{(1)}$ are independent, and the score at $\vartheta^\star$ can be written as

$$\dot{\ell}_{\text{test},T}(\vartheta^\star) = \begin{pmatrix} A^{(0)} + A^{(1)} \\ A^{(1)} \end{pmatrix}.$$

By the independent increments property of Poisson process [12], the vectors $Y_{01}, \ldots, Y_{0L_T^{(0)}}$ are i.i.d. with $\mathbb{E}(Y_{0k}) = \mathbb{E}(Y) = 0$ and $\mathrm{Var}(Y_{0k}) = \mathrm{Var}(Y) = M^\star$.

By the multivariate central limit theorem and Slutsky's theorem, we obtain

$$\frac{A^{(0)}}{\sqrt{T}} = \sqrt{\frac{L_T^{(0)}}{T}} \left(\frac{1}{\sqrt{L_T^{(0)}}} \sum_{k=1}^{L_T^{(0)}} Y_{0k} \right) \xrightarrow[T\to\infty]{d} \sqrt{s_i}\,\mathcal{N}_2(0, M^\star) = \mathcal{N}_2(0, s_i M^\star).$$

Similarly,

$$\frac{A^{(1)}}{\sqrt{T}} = \sqrt{\frac{L_T^{(1)}}{T}} \frac{A^{(1)}}{\sqrt{L_T^{(1)}}} \xrightarrow[T\to\infty]{d} \mathcal{N}_2(0, (1-s_i)M^\star).$$

Therefore,

$$Z_T := \frac{1}{\sqrt{T}} \dot{\ell}_{\mathrm{test},T}(\vartheta^\star) = \frac{1}{\sqrt{T}} \begin{pmatrix} A^{(0)} + A^{(1)} \\ A^{(1)} \end{pmatrix} \xrightarrow[T\to\infty]{d} \mathcal{N}_4(0, I^\star),$$

where

$$I^\star = \begin{pmatrix} M^\star & (1-s_i)M^\star \\ (1-s_i)M^\star & (1-s_i)M^\star \end{pmatrix}.$$

Step 3: convergence of the normalized Hessian and estimator.

We now show that the normalized Hessian converges to the information matrix. For the parameter $\vartheta = (b, \mu, \delta_b, \delta_\mu)^\top$, the Hessian of the test log-likelihood is

$$\ddot{\ell}_{\mathrm{test,T}}(\vartheta) = \begin{pmatrix} -\dfrac{N_{i,\mathrm{test}}^{(0)}}{b^2} - \dfrac{N_{i,\mathrm{test}}^{(1)}}{(b+\delta_b)^2} & 0 & -\dfrac{N_{i,\mathrm{test}}^{(1)}}{(b+\delta_b)^2} & 0 \\ 0 & -\dfrac{N_{i,\mathrm{test}}^{(0)}}{\mu^2} - \dfrac{N_{i,\mathrm{test}}^{(1)}}{(\mu+\delta_\mu)^2} & 0 & -\dfrac{N_{i,\mathrm{test}}^{(1)}}{(\mu+\delta_\mu)^2} \\ -\dfrac{N_{i,\mathrm{test}}^{(1)}}{(b+\delta_b)^2} & 0 & -\dfrac{N_{i,\mathrm{test}}^{(1)}}{(b+\delta_b)^2} & 0 \\ 0 & -\dfrac{N_{i,\mathrm{test}}^{(1)}}{(\mu+\delta_\mu)^2} & 0 & -\dfrac{N_{i,\mathrm{test}}^{(1)}}{(\mu+\delta_\mu)^2} \end{pmatrix}.$$

Evaluating this matrix at $\vartheta^\star = (b^\star, \mu^\star, 0, 0)^\top$ and using the limits

$$\frac{N_{i,\mathrm{test}}^{(0)}}{T} \xrightarrow[T\to\infty]{\mathbb{P}} b^\star s_i, \qquad \frac{N_{i,\mathrm{test}}^{(1)}}{T} \xrightarrow[T\to\infty]{\mathbb{P}} b^\star (1-s_i),$$

we obtain

$$-\frac{1}{T}\ddot{\ell}_{\text{test,T}}(\vartheta^\star) \xrightarrow[T\to\infty]{\mathbb{P}} I^\star,$$

By Steps 1–2 and the convergence of the normalized negative Hessian, the assumptions of the standard asymptotic normality theorem for maximum likelihood estimators are satisfied; see, for instance, [19]. Hence,

$$\sqrt{T}(\widehat{\vartheta}_T - \vartheta^\star) \xrightarrow[T\to\infty]{d} \mathcal{N}_4(0, (I^\star)^{-1}).$$

The same argument applies to the restricted model, obtained by imposing $\delta_b = \delta_\mu = 0$.

Step 4: identification of the restriction and quadratic form.

Let

$$h(\vartheta) = \begin{pmatrix} \delta_b \\ \delta_\mu \end{pmatrix} = H\vartheta, \quad \text{where} \quad H = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

So, the null hypothesis can be written as the restriction

$$\mathcal{H}_0^{(i)} \quad \Longleftrightarrow \quad h(\vartheta) = 0,$$

The matrix H has rank 2 and the likelihood-ratio statistic is

$$D_i^{\text{test}} = 2\left[\sup_{\vartheta\in\Theta} \ell_{\text{test},T}(\vartheta) - \sup_{\vartheta:\, h(\vartheta)=0} \ell_{\text{test},T}(\vartheta)\right].$$

Therefore, by the standard likelihood-ratio theorem, see [18], we obtain

$$D_i^{\text{test}} \xrightarrow[T\to\infty]{d} \chi_2^2.$$

This proves Proposition 1.

Appendix B

Proof of Proposition 2

We prove that the proposed monitoring procedure controls the probability of raising at least one false alarm over an infinite monitoring horizon.

For each $d \geq 1$, the likelihood ratio L_d is non-negative. Under the global pre-change null hypothesis, the block X_d has density f_0 and is independent of the past filtration $\mathcal{F}_{d-1}$. Therefore,

$$\mathbb{E}_{\mathcal{H}_0}(L_d \mid \mathcal{F}_{d-1}) = \mathbb{E}_{\mathcal{H}_0}(L_d) = \int \frac{f_1(x)}{f_0(x)} f_0(x)\, dx = \int f_1(x)\, dx = 1.$$

Since Y_{d-1} is $\mathcal{F}_{d-1}$-measurable and $Y_d = Y_{d-1}L_d$, we obtain

$$\mathbb{E}_{\mathcal{H}_0}(Y_d \mid \mathcal{F}_{d-1}) = \mathbb{E}_{\mathcal{H}_0}(Y_{d-1}L_d \mid \mathcal{F}_{d-1}) = Y_{d-1}\mathbb{E}_{\mathcal{H}_0}(L_d \mid \mathcal{F}_{d-1}) = Y_{d-1}.$$

Thus, $(Y_d)_{d\geq 0}$ is a non-negative martingale under $\mathbb{P}_{\mathcal{H}_0}$, with $Y_0 = 1$.

By Ville's inequality, for any threshold $c > 0$,

$$\mathbb{P}_{\mathcal{H}_0}\left(\sup_{d\geq 0} Y_d \geq c\right) \leq \frac{\mathbb{E}_{\mathcal{H}_0}(Y_0)}{c} = \frac{1}{c}.$$

Taking $c = 1/\alpha$, we get

$$\mathbb{P}_{\mathcal{H}_0}\left(\sup_{d\geq 0} Y_d \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Moreover, by definition of the alarm time A,

$$\{A < +\infty\} = \left\{\sup_{d\geq 1} Y_d \geq \frac{1}{\alpha}\right\}.$$

Hence,

$$\mathbb{P}_{\mathcal{H}_0}(A < +\infty) \leq \alpha.$$

This proves Proposition 2.

Appendix C

Notation and Conventions

- We use the convention $Z_A^{(C)}(B)$ to indicate that the quantity Z is computed with index set A, over the time interval B, and in state C. The state $C = 0$ corresponds to the pre-change state (before the change-point), whereas $C = 1$ corresponds to the post-change state (after the change-point). If one of the three components A, B, or C is omitted, the omitted component is taken over its whole domain, and the quantity is computed conditional on the remaining specified components. The pre-change state corresponds to events such that $T_{ij} \leq S_i$, while the post-change state corresponds to events such that $T_{ij} > S_i$. For example, $V_{ir}^{(0)}((0, S_i])$ is the sum of marks for ewe i and ram r up to and including the candidate change-point S, whereas $V_{ir}^{(1)}((S_i, T])$ is the corresponding sum strictly after S. Since the intervals $(0, S_i]$ and $(S_i, T]$ already identify the pre-change and post-change periods, respectively, these quantities may equivalently be written as $V_{ir}((0, S_i])$ and $V_{ir}((S_i, T])$ or as $V_{ir}^{(0)}$ and $V_{ir}^{(1)}$.

- A true positive (TP) means $S_i < T$ and the algorithm detects a change-point. A false positive (FP) means $S_i = T$ but the algorithm detects a change-point. A true negative (TN) means $S_i = T$ and no change-point is detected. A false negative (FN) means $S_i < T$ but the algorithm does not detect a change-point. When $S_i < T$, we define $e_i = \widehat{S}_i - S_i$. The mean absolute error is the mean of $|e_i|$, and the variance of the error is the empirical variance of e_i. For online methods, if A_i denotes the alarm time, the alarm delay is $A_i - S_i$.

- For the event-level comparison with Nathalie's empirical manual rule, we use the labels `OKKO`, `OK?`, `OKOK`, `KOKO`, `KO?`, and `KOOK`. The first two letters correspond to the offline algorithm decision: `OK` means that

the offline algorithm detects a rupture, and KO means that the offline algorithm does not detect a rupture. The last part corresponds to Nathalie's manual rule based on the cumulative number of Alpha-D readings within the day: OK means at least 6 readings, KO means fewer than 3 readings, and ? means between 3 and 5 readings. Therefore, OKOK means that both the offline algorithm and Nathalie's manual rule detect heat, KOKO means that neither detects heat, OKKO means that the offline algorithm detects a rupture but Nathalie's rule does not detect heat, and KOOK means that Nathalie's rule detects heat but the offline algorithm does not detect a rupture. The labels OK? and KO? correspond to uncertain manual cases. Since this classification is applied to each recorded event, the resulting counts are event-level counts and not ewe-level counts.

Symbol	Meaning
n	Number of ewes, or the number of individual marked Poisson processes observed.
i	Index of a ewe, with $i \in \{1, \ldots, n\}$.
j	Index of an event of ewe i.
R	Number of rams in the dataset.
r	Index of a ram, with $r \in \{1, \ldots, R\}$.
$\mathcal{R}$	Set of ram indices, defined by $\mathcal{R} = \{1, \ldots, R\}$.
$T_A^{(C)}(B)$	Collection of event times associated with the index set A, the time interval B, and the state C.
$U_A^{(C)}(B)$	Collection of marks associated with the index set A, the time interval B, and the state C.
$R_A^{(C)}(B)$	Collection of ram labels associated with the index set A, the time interval B, and the state C.
$X_A^{(C)}(B)$	Observed events, $X_A^{(C)}(B) = \big(T_A^{(C)}(B), U_A^{(C)}(B), R_A^{(C)}(B)\big)$.
$N_A^{(C)}(B)$	Number of events in $X_A^{(C)}(B)$.
$V_A^{(C)}(B)$	Sum of marks associated with $X_A^{(C)}(B)$.
$\mathcal{G}_A^{(C)}(B)$	Set of event times $\{T_A^{(C)}(B)\}$.
S_i	Change-point of ewe i. $S_i = T$ means there is no change-point.
$\widehat{S}_i$	Estimated change-point of S_i.
$\lambda_A = (\lambda_A^{(0)}, \lambda_A^{(1)})$	Event intensity in the pre-change and post-change states.
$\mu_A = (\mu_A^{(0)}, \mu_A^{(1)})$	Exponential rate parameter of the marks in the pre-change and post-change states.
$\ell_i(S; \theta)$	Log-likelihood contribution of ewe i for a candidate change-point S and parameter vector θ.
$\ell(\mathbf{S}, \theta)$	Total log-likelihood over all ewes, where $\mathbf{S} = (S_1, \ldots, S_n)$.

Table C.1: List of Notation